

# Wangyang Ying

Computer Science, Ph.D. Candidate  
School of Computing and AI  
Arizona State University

Mobile: +1 (689) 275-4229  
Email: [yingwangyang@gmail.com](mailto:yingwangyang@gmail.com)  
Web Page: <https://yingwangyang.github.io>

## Summary

I am a Ph.D. candidate in Computer Science with hands-on industrial experience in **large-scale search and recommendation** systems at Tencent and Alibaba, and active research in **LLM reasoning and agent systems**. My work focuses on **representation learning, data-centric modeling, and system-level LLM techniques**, with an emphasis on efficiency, reliability, and deployability. I aim to pursue Applied Scientist / Research Scientist / Machine Learning Engineer roles where research ideas can be translated into production-grade AI systems.

## Education

Ph.D., Computer Science, Arizona State University, 2023 - 2026  
M.S., Computer Science, Sichuan University, 2016 - 2019  
B.E, Computer Science, Sichuan University, 2012 - 2016

## Research & Industry Experience

### *(1) Research: LLM Reasoning, Agents, and Representation Learning*

#### LLM Reasoning

Focused on improving the **accuracy, stability, and controllability of multi-step reasoning** in large language models, with emphasis on post-training optimization and efficiency under constrained settings.

- *Developed latent-space reasoning and inference-time refinement methods to correct reasoning trajectories without modifying model parameters, enabling efficient post-training improvements.*
- *Investigated feedback- and memory-based mechanisms to reduce error accumulation and stabilize reasoning across long chains of inference.*
- *Evaluated methods on multiple multi-step reasoning benchmarks, demonstrating consistent accuracy and robustness gains under limited compute budgets.*

#### LLM Agent Systems

Studied **agent-based frameworks** for iterative reasoning and structured generation in LLMs, with emphasis on **feedback, interaction, and limited supervision**.

- *Designed multi-agent coordination mechanisms using feedback loops to support iterative refinement beyond single-pass generation.*
- *Explored in-context learning-driven agent behaviors to improve robustness, efficiency, and generalization across reasoning and structured transformation tasks.*
- *Analyzed trade-offs between agent interaction depth, performance gains, and inference cost.*

#### Representation Learning

Focused on **data-centric** and **representation-driven** methods that improve model behavior without relying on model scaling.

- *Reformulated discrete search problems in feature engineering as generative learning tasks, enabling structured learning from noisy or weakly supervised data in retrieval, ranking, and downstream prediction settings.*
- *Developed robust representation learning techniques that improve generalization and stability under real-world constraints such as limited labels and distribution shift.*
- *Validated approaches on benchmark and applied tasks, showing strong performance with lightweight models.*

## **(2) Internship: LLM Agent, RAG, and Reliability**

**Internship**, NEC Laboratories America - Research Intern, May 2025 - Aug 2025

- *Developed multi-agent LLM frameworks for structured knowledge extraction, producing procedural graph representations to support retrieval-augmented generation (RAG) pipelines.*
- *Designed structured representations that improve grounding and retrieval consistency, enabling more reliable downstream reasoning and generation.*
- *Explored mapping user-specific workflows into structured knowledge to support personalization in LLM-powered retrieval pipelines.*

**Internship**, A\*STAR Singapore - Research Intern, Jun 2024 - Aug 2024

- *Studied reliability and trustworthiness of LLM-based systems in medical and high-stakes domains.*
- *Conducted systematic analysis of failure modes and jailbreak strategies, providing insights applicable to improving robustness and safety in production LLM and retrieval-augmented systems.*

## **(3) Industry: Large-Scale Search and Recommendation Systems**

**Full Time**, Tencent - Machine Learning Engineer, Nov 2020 - Aug 2022

- **Time-Sensitive Query Understanding and Ranking Control.**
  - *Trained a semantic model to identify temporal intent from query and generate time-sensitivity signals.*
  - *Integrated signals into ranking strategies to dynamically balance relevance and freshness.*
  - *Enabled product-level content aggregation cards for strongly time-sensitive queries, improving user engagement and interaction.*
- **Event Aggregation and Burst Detection for News Search.**
  - *Developed title-level event extraction models to identify event semantics and burst signals.*
  - *Clustered news articles into event-level groups for event-aware retrieval and de-duplication.*
  - *Released a benchmark dataset for event extraction and evaluation (EMNLP).*
- **Retrieval System Integration and Reliability Improvement.**
  - *Integrated new content streams into the production retrieval pipeline and validated coverage and stability under real traffic.*

**Full Time**, Alibaba - Machine Learning Engineer, Jun 2019 - Oct 2020

- **Tagging System Implementation.**
  - *Built a recommendation-oriented hierarchical tagging system for short-form videos, addressing cold-start and retrieval coverage issues in recommendation pipelines.*
- **Online Impact & Validation for the Tagging System.**
  - *Integrated tags into retrieval pipelines via tag-based matching and embedding retrieval, validating system impact through improvements in CTR, watch time, and retention.*

## **Selected Publications**

I have published papers in top-tier venues, including KDD, NeurIPS, EMNLP, AAAI, CIKM, IJCAI, and TKDD. A complete list of publications is available on my personal website.

1. [TKDD] Feature Selection as Deep Sequential Generative Learning.
  - *Formulates feature selection as a sequential generative process, enabling structured and controllable selection of informative features for downstream prediction tasks.*
2. [KDD] Unsupervised Generative Feature Transformation via Graph Contrastive Pre-training and Multi-objective Fine-tuning.

- *Transforms features into simpler and more discriminative representation spaces, enabling strong downstream performance with lightweight models.*
3. [AAAI] Efficient Post-Training Refinement of Latent Reasoning in Large Language Models.
    - *Proposed a training-free, post-training approach to improve the stability and accuracy of multi-step reasoning in large language models by refining latent reasoning trajectories.*
  4. [IJCAI] Unsupervised Feature Transformation via In-context Generation, Generator-critic LLM Agents, and Duet-play Teaming.
    - *Introduced an agent-based, generative formulation for discrete search and feature transformation, enabling effective learning from unlabeled data through iterative feedback.*
  5. [CIKM] Revolutionizing Biomarker Discovery: Leveraging Generative AI for Bio-Knowledge-Embedded Continuous Space Exploration.
    - *Applied generative feature selection to high-dimensional biological data to identify a compact set of informative features, improving disease prediction performance.*
  6. [EMNLP] Title2Event: Benchmarking Open Event Extraction with a Large-scale Chinese Title Dataset.
    - *Event extraction from short news titles, enabling event-level representation, aggregation, and event-driven news search.*

## **Honors**

- KDD 2024 Travel Award
- National Scholarship, Top 1%, 2019
- Postgraduate Scholarship in Sichuan University, 1st Prize, Top 1%, 2018, 2019

## **Professional Activities and Services**

- Neural Information Processing Systems (**NeurIPS**) 2025
- International Conference on Machine Learning (**ICML**) 2025
- The International Conference on Learning Representations (**ICLR**) 2024, 2025, 2026
- The Conference on Knowledge Discovery and Data Mining (**KDD**) 2024, 2025
- The Association for the Advancement of Artificial Intelligence (**AAAI**) 2025
- The Conference on Information and Knowledge Management (**CIKM**) 2023, 2024
- The IEEE International Conference on Big Data (**BigData**) 2023, 2024, 2025
- ACM Transactions on Knowledge Discovery from Data (**TKDD**)
- IEEE Transactions on Knowledge and Data Engineering (**TKDE**)
- IEEE Transactions on Cognitive and Developmental Systems (**TCDS**)
- IEEE Transactions on Big Data (**TBD**)